

Lab_Assignment_21.2

AI Assisted Coding B37

2303A52077

Task 1 – AI-Assisted Pair Programming

PROMPT

#Generate a simple Python program for a To-Do List Manager.

#Enhance this To-Do List program by adding error handling, delete option, and better structure

CODE

```
#Generate a simple Python program for a To-Do List Manager.
tasks = []
while True:
    print("\n1. Add Task")
    print("2. View Tasks")
    print("3. Exit")
    choice = input("Enter choice: ")
    if choice == '1':
        task = input("Enter task: ")
        tasks.append(task)
    elif choice == '2':
        for t in tasks:
            print(t)
    elif choice == '3':
        break
    else:
        print("Invalid choice")
```

```

#Enhance this To-Do List program by adding error handling, delete option, and better structure
tasks = []

def add_task():
    task = input("Enter task: ").strip()
    if not task:
        print("Task cannot be empty!")
        return
    tasks.append(task)
    print("Task added!")

def view_tasks():
    if not tasks:
        print("No tasks available.")
    else:
        for i, t in enumerate(tasks, 1):
            print(f"{i}. {t}")

def delete_task():
    view_tasks()
    try:
        index = int(input("Enter task number to delete: "))
        if 1 <= index <= len(tasks):
            removed = tasks.pop(index - 1)
            print(f"Deleted: {removed}")
        else:
            print("Invalid task number")
    except ValueError:
        print("Please enter a valid number")

while True:
    print("\n1. Add Task")
    print("2. View Tasks")
    print("3. Delete Task")
    print("4. Exit")

    choice = input("Enter choice: ")

    if choice == '1':
        add_task()
    elif choice == '2':
        view_tasks()
    elif choice == '3':
        delete_task()
    elif choice == '4':
        break
    else:
        print("Invalid choice")

```

OUTPUT

```
1. Add Task
2. View Tasks
3. Exit
Enter choice: 1
Enter task: WATCH IPL

1. Add Task
2. View Tasks
3. Exit
Enter choice: 2
WATCH IPL

1. Add Task
2. View Tasks
3. Exit
Enter choice: 3

1. Add Task
2. View Tasks
3. Delete Task
4. Exit
Enter choice: 4
```

EXPLANATION

The initial program was a basic To-Do list that could only add and view tasks. It did not have error handling or additional features like deleting tasks. Using AI, we improved the program by adding input validation, delete functionality, and better structure using functions. The updated version is more user-friendly and reliable. This shows how AI helps in improving code quality and development.

Task 2 – Version Control Integration

CODE

```
#calculator.py
def add(x, y):
    return x + y
def subtract(x, y):
    return x - y
def multiply(x, y):
    return x * y
def divide(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x / y
def power(x, y):
    return x ** y
def modulus(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x % y
def floor_division(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x // y
def main():
    try:
        x = float(input("Enter first number: "))
        y = float(input("Enter second number: "))
        print("\nChoose operation:")
        print("1. Add")
        print("2. Subtract")
        print("3. Multiply")
        print("4. Divide")
        print("5. Power")
        print("6. Modulus")
        print("7. Floor Division")
        choice = input("Enter choice (1-7): ")
```

OUTPUT

```
Enter choice (1-7): 1
Result: 46.0
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git add .
>> git commit -m "Initial calculator program"
[ai-enhancements 87752b4] Initial calculator program
1 file changed, 54 insertions(+)
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git checkout -b ai-enhancements
fatal: a branch named 'ai-enhancements' already exists
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git checkout ai-enhancements
Already on 'ai-enhancements'
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git branch -m main
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> █
```

EXPLANATION

We initialized a Git repository and created a branch for AI-assisted changes. The calculator program was developed and improved using AI suggestions. The changes were committed and the branch was finalized as the main branch. This demonstrates how version control supports development with AI assistance.

Task 3 – AI for Commit Message Generation

PROMPT

Generate a Git commit message for: Added menu-driven calculator ded user input Added error handling

CODE

```
#calculator python
def add(x, y):
    return x + y
def subtract(x, y):
    return x - y
def multiply(x, y):
    return x * y
def divide(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x / y
def power(x, y):
    return x ** y
def modulus(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x % y
def floor_division(x, y):
    if y == 0:
        return "Error: Division by zero"
    return x // y
def main():
    try:
        x = float(input("Enter first number: "))
        y = float(input("Enter second number: "))
        print("\nChoose operation:")
        print("1. Add")
        print("2. Subtract")
        print("3. Multiply")
        print("4. Divide")
        print("5. Power")
        print("6. Modulus")
        print("7. Floor Division")
        choice = input("Enter choice (1-7): ")
    except ValueError:
        print("Invalid input. Please enter a valid number.")

if __name__ == "__main__":
    main()
```

ai 21.2 > 3.py > ...

```
1 #Generate a Git commit message for: Added menu-driven calculator ded user input Added error handling
2 Added a menu-driven calculator that takes user input for two numbers
3 and allows the user to choose an operation
4 (addition, subtraction, multiplication, division, power, modulus, floor division).
5 Implemented error handling to manage invalid inputs and division by zero cases.
6
7
```

OUTPUT

```
PS C:\Users\gauti\OneDrive\Documents\SEM-6\AI\docs\codes> git log
commit 87752b4f842963746b3cb2049e5b683400c9e927 (HEAD -> main)
Author: 2303A52074 <2303A52074@sru.edu.in>
Date: Tue Mar 31 14:54:47 2026 +0530

    Initial calculator program

commit 999a7fa2f3fa08e5e75f1e205e75cc64baae9e8f
Author: 2303A52074 <2303A52074@sru.edu.in>
Date: Tue Mar 31 14:48:07 2026 +0530

    Added AI-assisted improvements
```

EXPLANATION

AI was used to generate clear and meaningful commit messages based on code changes.

Human-written messages are often short and unclear, while AI-generated messages are more descriptive.

This helps in better understanding of changes in the project history.

Thus, AI improves documentation and traceability in version control systems.

Task 4 – Pair Programming Simulation

PROMPT

Improve this calculator program by adding user input, menu options, and error handling.

CODE

```
ai 21.2 > 4.py > ...
1  def main():
2      try:
3          x = float(input("Enter first number: "))
4          y = float(input("Enter second number: "))
5          print("1. Add\n2. Subtract")
6          choice = input("Enter choice: ")
7
8          if choice == '1':
9              print("Result:", x + y)
10             elif choice == '2':
11                 print("Result:", x - y)
12             else:
13                 print("Invalid choice")
14
15         except ValueError:
16             print("Invalid input")
17     if __name__ == "__main__":
18         main()
19
```

OUTPUT

```

PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git init
>> git add .
>> git commit -m "Initial calculator program"
Reinitialized existing Git repository in C:/Users/gauta/OneDrive/Documents/SEM-6/AI/docs/codes/.git
[main 35db6f3] Initial calculator program
 2 files changed, 25 insertions(+)
 create mode 100644 ai 21.2/3.py
 create mode 100644 ai 21.2/4.py
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git checkout -b ai-enhancements
Switched to a new branch 'ai-enhancements'
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git add .
>> git commit -m "Added user input and error handling"
On branch ai-enhancements
nothing to commit, working tree clean
PS C:\Users\gauta\OneDrive\Documents\SEM-6\AI\docs\codes> git checkout main
>> git merge ai-enhancements
Switched to branch 'main'
Already up to date.

```

EXPLANATION

Student A created the initial version of the program.

Student B used AI to improve the code by adding features and error handling.

Both versions were managed using Git branches.

Finally, the changes were merged to complete the program collaboratively.